

Berker Demirel

Personal Webpage
Google Scholar

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RESEARCH INTERESTS

My research focuses on representation learning, particularly on the geometry and structure of neural representations. I am interested in how representations are organized, when they collapse and how they behave under distribution shift. More broadly, my work includes self-supervised learning, generalization and generative modeling.

EDUCATION

- **Institute of Science and Technology Austria (ISTA)** Klosterneuburg, Austria
PhD Candidate in Computer Science Sep 2023 – Present
Research: Representation learning, self-supervised learning, representation geometry, and generative modeling
Supervisor: Francesco Locatello
- **Sabanci University** Istanbul, Turkey
MSc in Computer Science; GPA: 3.91 Sep 2021 – Jun 2023
Research: Domain generalization and group activity recognition
Supervisor: Huseyin Ozkan
- **Sabanci University** Istanbul, Turkey
BS Computer Science and Engineering, BA Economics (Double Major) Sep 2015 – Jun 2021
Mathematics Minor; GPA: 3.84

SELECTED PUBLICATIONS

- **λ -JEPA: Spectral Anti-Collapse Regularization for Self-Supervised Learning**
Berker Demirel, Cl  mentine Domin  , Valentino Maiorca, Marco Fumero, Marco Mondelli, Francesco Locatello.
arXiv preprint arXiv:2609.35288, 2026.
- **Assessing Sample Quality in Conditional Generation under Compositional Shift**
Berker Demirel, Valentino Maiorca, Marco Fumero, Theofanis Karaletsos, Francesco Locatello.
Advances in Neural Information Processing Systems (NeurIPS), 2026.
- **MorphGen: Controllable and Morphologically Plausible Generative Cell-Imaging**
Berker Demirel, Marco Fumero, Theofanis Karaletsos, Francesco Locatello.
Advances in Neural Information Processing Systems (NeurIPS), 2026.
- **Out-of-Distribution Detection with Relative Angles**
Berker Demirel, Marco Fumero, Francesco Locatello.
Advances in Neural Information Processing Systems (NeurIPS), 2025.
- **DECOMPL: Decompositional Learning with Attention Pooling for Group Activity Recognition from a Single Volleyball Image**
Berker Demirel, Huseyin Ozkan.
IEEE International Conference on Image Processing (ICIP), 2024.

COMMUNITY & SERVICE

- **Reviewer:** ICLR, NeurIPS, TMLR, UniReps, NeurReps, CauScien.

HONORS AND AWARDS

- Ranked 1st among a cohort of 20 students in the Department of Economics at Sabanci University.
- Ranked 5th among a cohort of 90 students in the Department of Computer Science & Engineering at Sabanci University.
- Honor scholarship for five years, awarded after ranking 1,878th among approximately 2 million participants in the nationwide university admission examination.